

Final Project EDT

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1 Phase 1: Problem Definition and Data Acquisition

1.1 Event Definition and Study Area

This project revisits the case study analyzed in the Midterm Project: the collapse of the Kakhovka Dam on June 6, 2023, located on the Dnieper River in Kherson Oblast, southern Ukraine. The event caused catastrophic downstream flooding and the rapid drainage of the Kakhovka Reservoir, making it one of the most significant hydrological disasters in recent European history.

The Region of Interest (ROI) is centered on the area surrounding Nova Kakhovka, where the effects of reservoir drainage were most clearly observable in satellite imagery. This ROI was selected based on image availability and the visibility of hydrological changes between the pre- and post-event dates.

1.2 Satellite Imagery Metadata

Two Sentinel-2 Level-2A images were acquired from the Copernicus Open Access Hub. Table 1 summarizes their main technical characteristics.

Table 1: Technical characteristics of the satellite imagery used.

Date	Satellite	Level	Resolution	Cloud Cover	Role
June 18, 2023	Sentinel-2 A/B	Level-2A	10 m	<30%	Post-event

Level-2A products provide Bottom-of-Atmosphere (BOA) reflectance, eliminating the need for additional atmospheric correction. The bands used for this project span from the visible to the shortwave infrared spectrum, as detailed in Table 2.

Table 2: Sentinel-2 spectral bands used for dataset construction.

Band	Name	Wavelength (μm)
B2	Blue	0.490
B3	Green	0.560
B4	Red	0.665
B5	Red Edge 1	0.705
B6	Red Edge 2	0.740
B7	Red Edge 3	0.783
B8	Near-Infrared (NIR)	0.842
B8A	Narrow NIR	0.865
B11	SWIR 1	1.610
B12	SWIR 2	2.190

1.3 Dataset Construction

To prepare the training dataset for the machine learning phase, pixel values were extracted from the Sentinel-2 band stack at the locations defined by the training polygons digitized in QGIS. Each polygon corresponds to one of four land cover classes: Water, Vegetation, Bare Soil, and Built-up.

The extraction was performed using a Python script based on the **rasterio** and **geopandas** libraries. For each pixel falling within a training polygon, the script records its geographic coordinates (Latitude, Longitude), the reflectance values for all ten spectral bands, and the corresponding class label. The output was exported as a Tab-Separated Values (TSV) file, as required for the subsequent training phase.

Table 3 illustrates the structure of the resulting dataset.

Table 3: Structure of the exported TSV training dataset.

Column	Type	Description
Latitude	Float	Geographic latitude of the pixel centroid
Longitude	Float	Geographic longitude of the pixel centroid
B2	Float	Blue band reflectance
B3	Float	Green band reflectance
B4	Float	Red band reflectance
B5	Float	Red Edge 1 reflectance
B6	Float	Red Edge 2 reflectance
B7	Float	Red Edge 3 reflectance
B8	Float	NIR band reflectance
B8A	Float	Narrow NIR reflectance
B11	Float	SWIR 1 reflectance
B12	Float	SWIR 2 reflectance
label	String	Land cover class (Water, Vegetation, Bare Soil, Built-up)